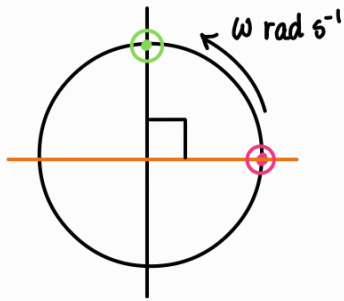


Physical occurrence of a sinusoid —

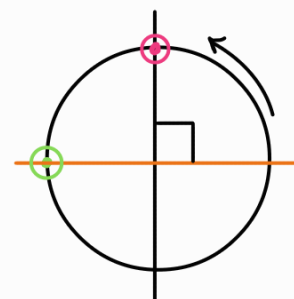
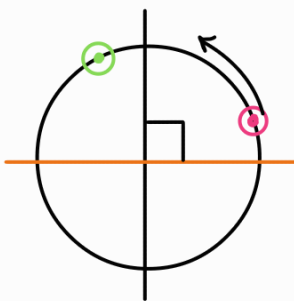
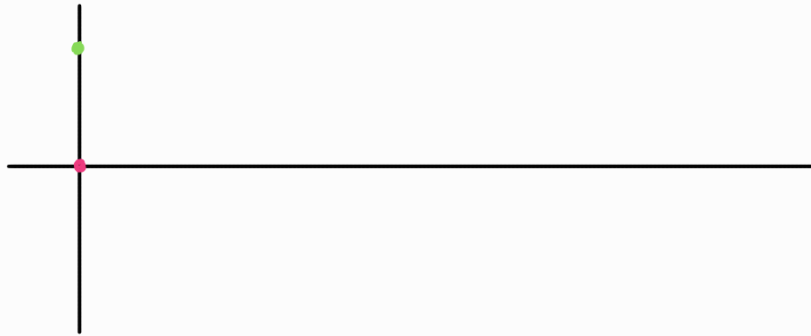
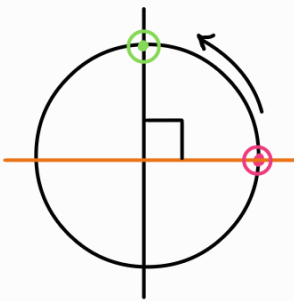
We have learned sine and cosine waves multiple times throughout our academic life. But have we ever asked ourselves where these sines and cosines actually occur in real life?

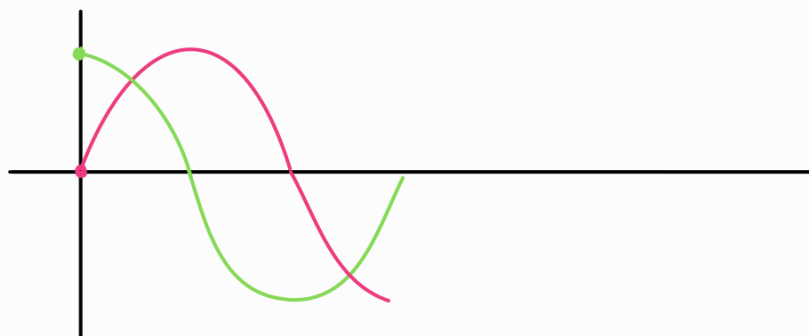
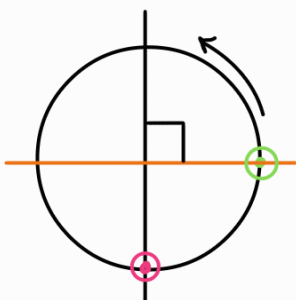
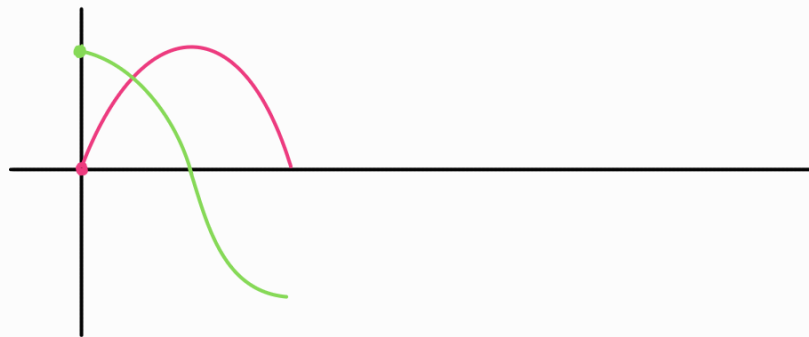
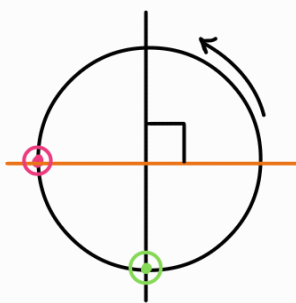
Imagine a car's wheel —



The red and the green dots represent two sensors that collect position data over time. Both the dots collect position data with respect to the orange line. Now if we plot the position data over time, let's see what happens.

In most books we measure wrt the y-axis for consistency with complex sinusoids.





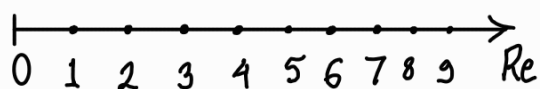
Do you see where this is going. Any circular motion can be decomposed to sine and cosine. Pretty neat, no? The frequency makes sense now. The number of rotation per second is the sinusoid frequency. The amplitude is the radius of the wheel.

Question — What would happen if the wheel rotated clockwise?

Answer — Negative frequency.

e^{at} and its physical implications —

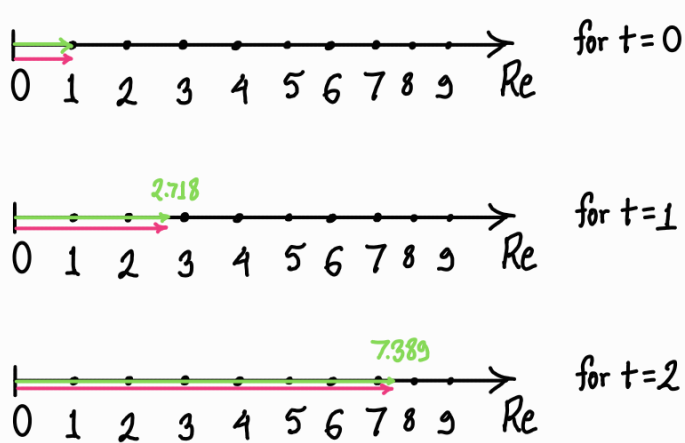
Assume we are in a 1D vector space. Think of the number line.



Let $x(t) = e^t$ indicate any point on the real number line. Assume it is the displacement vector.

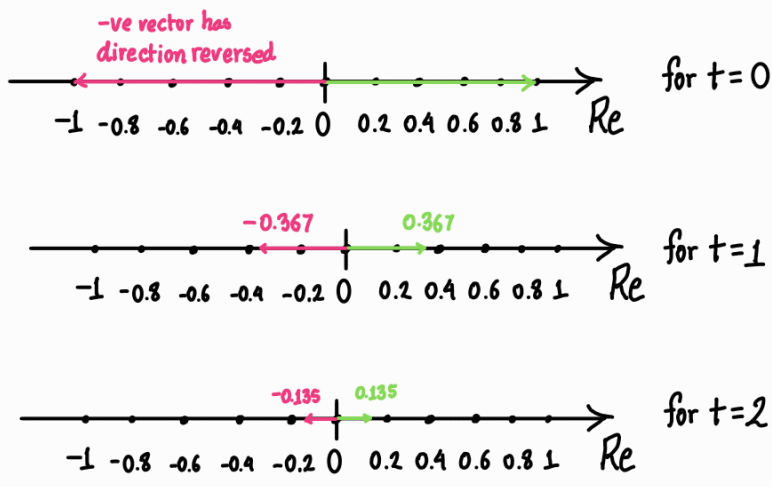
Then the rate of change, $v(t) = \frac{d}{dt} e^t = e^t$ is the velocity vector. The displacement follows the velocity's direction, and higher the velocity, faster the change of displacement.

So if we start visualizing the system —



displacement speeds up since velocity amps up
the displacement also updates in the direction of the velocity

Now let $x(t) = e^{-t}$. Then $v(t) = -e^{-t}$. The system then becomes —

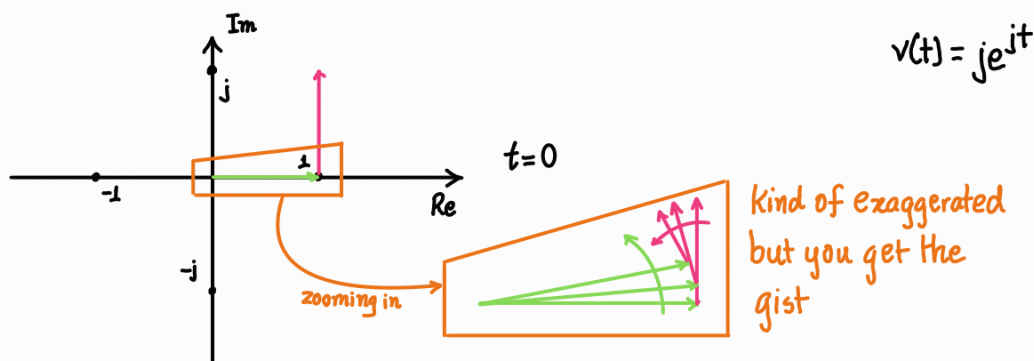


the displacement decreases overtime because velocity
is opposing the displacement direction
the change is also slowing down as velocity magnitude
is decreasing

for $x(t) = e^{at}$ the effect of increase or decrease speeds up.

But what happens when $x(t) = e^{jt}$ where $j = \sqrt{-1}$?

So in case you don't remember, the imaginary axis is perpendicular to the real axis. So this is what will happen —



$$v(t) = je^{jt}$$

Kind of exaggerated
but you get the
gist

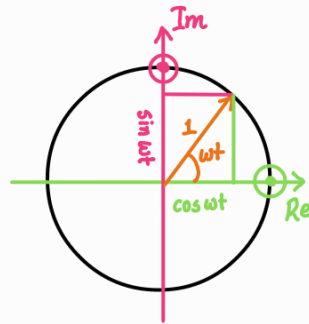
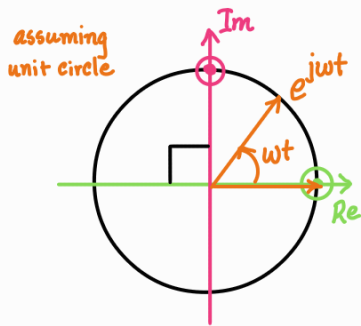
$x(t) = e^{jt}$ forms a circle while moving counterclockwise

$x(t) = e^{-jt}$ moves clockwise

I leave it upto you to figure out how long it takes for $x(t)$ to complete 1 rotation. Plug in $t = \frac{\pi}{2}, \pi, \frac{3\pi}{4}$ & 2π

We can control the rotation speed by adding a ω term. So effectively $z(t) = e^{j\omega t}$ for anticlockwise circular movement and $z(t) = e^{-j\omega t}$ for clockwise movement. These complex exponentials now have an angular frequency of ω rad s^{-1} .

Reformulating the wheel problem—



$$e^{j\omega t} = \cos \omega t + j \sin \omega t \quad [\text{Euler's formula}]$$

Rotated for consistency

$e^{j\omega t}$ is called the complex sinusoid and it contains both sine and cosine info. The best part is that it keeps the sine and cosine terms independent.

In its most general form $x(t) = Ae^{j\omega t + j\phi} = Ae^{j\phi}e^{j\omega t}$

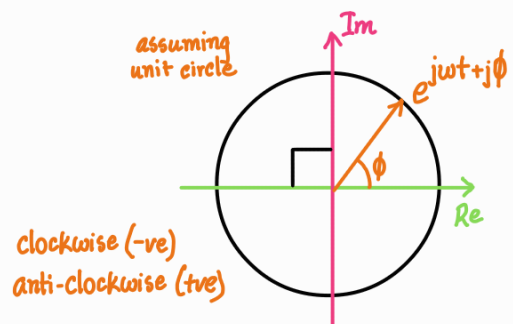
amplitude
frequency
phase (no counter intuitive shift here)

Complex sinusoids are periodic in CT.

$$x[n] = Ae^{j\phi}e^{j\Omega n}$$

$$\Omega = 2\pi \frac{f}{f_s}$$

$x[n]$ is periodic iff f_{norm} is rational.



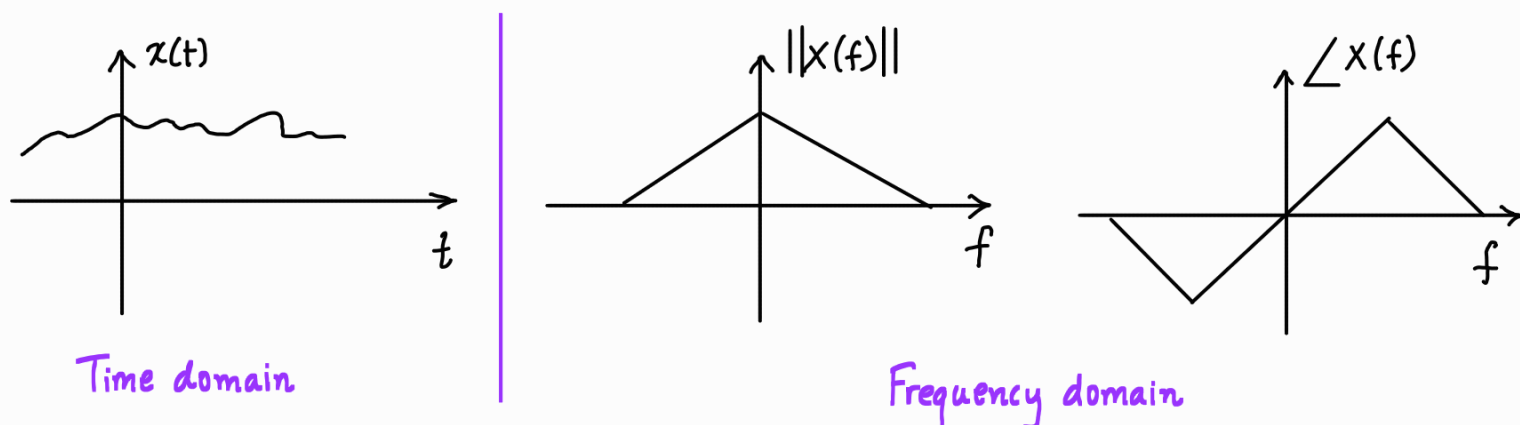
It is to be noted that $\|e^{j(\omega t + \phi)}\| = 1$ and $\angle e^{j(\omega t + \phi)} = \phi$

Also note for any complex number, $\text{magnitude} = \sqrt{\text{Re}^2 + \text{Im}^2}$ and $\phi = \tan^{-1} \frac{\text{Im}}{\text{Re}}$

We will be using complex sinusoids extensively from the next lecture.

Preparing for the frequency domain —

In general cases we would want our frequency domain to give us the magnitude and the initial phase value per frequency. So effectively,



So you can already sense where this is going. $X(f)$ in an ideal world will be a complex number. Also you may ask as to why we spent so much time on sinusoids. Fourier transformations break every signal into sinusoids. In the next lecture we will start with the trigonometric formulation of Fourier series and eventually will work towards the complex sinusoid formulation.

Orthogonal signals —

Signals that are mutually independent are called orthogonal signals. If 2 signals are orthogonal, they don't interfere with the other when passing through a common channel. At the end of the transmission, they can be separated without distortion.

Signal orthogonality follows linear algebra's orthogonality criteria. Consider you have 2 signals — $x_1[n]$ and $x_2[n]$. Then $x_1[n]$ and $x_2[n]$ are orthogonal iff

$$\sum_{n=-\infty}^{\infty} x_1[n] \cdot x_2[n] = 0 \quad \text{for aperiodic cases}$$

$$\sum_{n=1}^N x_1[n] \cdot x_2[n] = 0 \quad \text{for periodic cases}$$

for CT signals, the summation becomes an integration —

$$\int_{-\infty}^{\infty} x_1(t) \cdot x_2(t) dt = 0 \text{ for aperiodic cases}$$

$$\int_0^T x_1(t) \cdot x_2(t) dt = 0 \text{ for periodic cases}$$

If $x_1(t) = \cos \omega t$ and $x_2(t) = \sin \omega t$, are they orthogonal?

$$\begin{aligned} \int_0^T x_1(t) \cdot x_2(t) dt &= \int_0^T \frac{1}{2} (e^{j\omega t} + e^{-j\omega t}) \cdot \frac{1}{2j} (e^{j\omega t} - e^{-j\omega t}) dt \\ &= \frac{1}{4j} \int_0^T (e^{2j\omega t} - e^{-2j\omega t}) dt \\ &= \frac{1}{4j} \int_0^T 2 \sin 2\omega t dt \\ &= 0 \end{aligned}$$

So $x_1(t)$ and $x_2(t)$ are orthogonal.

If $x_1(t) = \sin \omega t$ & $x_2(t) = \sin(\omega t + 45^\circ)$, are they orthogonal?

$$\begin{aligned} \int_0^T x_1(t) \cdot x_2(t) dt &= \int_0^T \sin \omega t \cdot \sin(\omega t + 45^\circ) dt \\ &= \int_0^T \sin \omega t \cdot (\sin 45^\circ \cos \omega t + \cos 45^\circ \sin \omega t) dt \\ &= \frac{1}{\sqrt{2}} \int_0^T (\sin \omega t \cos \omega t + \sin^2 \omega t) dt \\ &= \frac{1}{\sqrt{2}} \int_0^T \frac{1}{2} \sin \frac{\omega t}{2} dt + \frac{1}{\sqrt{2}} \int_0^T \left(\frac{1}{2j} \right)^2 (e^{j\omega t} - e^{-j\omega t})^2 dt \\ &= \frac{-1}{4\sqrt{2}} \int_0^T (e^{j2\omega t} + e^{-j2\omega t} - 2) dt \end{aligned}$$

$$= \frac{-1}{4\sqrt{2}} \int_0^T (2\cos 2\omega t - 2) dt$$

$$= \frac{T}{2\sqrt{2}} \neq 0$$

Takeaway — For the same frequency sine and cosine are orthogonal. Anything in the middle are not

If $x_1(t) = \sin \omega t$ and $x_2(t) = \sin 2\omega t$, are they orthogonal?

$$\int_0^T x_1(t) \cdot x_2(t) dt = \int_0^T \sin \omega t \cdot \sin 2\omega t dt$$

$$= \int_0^T \frac{-1}{4} (e^{j\omega t} - e^{-j\omega t}) \cdot (e^{j2\omega t} - e^{-j2\omega t}) dt$$

$$= -\frac{1}{4} \int_0^T (e^{j3\omega t} - e^{j\omega t} - e^{-j\omega t} + e^{-j3\omega t}) dt$$

$$= -\frac{1}{4} \int_0^T (2\cos 3\omega t - 2\cos \omega t) dt = 0$$

Takeaway — Harmonics of a signal are orthogonal to the original signal.

Harmonics —

A regular signal is composed of a lot of different frequencies. The smallest non-zero frequency is the fundamental frequency (f_0). This concept mainly holds for periodic signals. Any component with frequency equal to $a f_0$ where $a \in \mathbb{N} - \{1\}$ is called a harmonic of the fundamental component.

$$x(t) = 5 + 13 \sin \omega t - 7 \cos 3\omega t + 5 \sin 3\omega t - 3 \cos 7\omega t + \sin 13\omega t$$

3rd harmonics
13th harmonic

fundamental component
7th harmonic

For DT signals however, the fundamental frequency is strictly given by $\Omega_0 = \frac{2\pi}{N}$

for example, $x[n] = 2\sin(2\pi \frac{11}{43} n)$ has $N=43$. So, $\Omega_0 = \frac{2\pi}{43}$. Then $2\pi \frac{11}{43}$ represents the 11th harmonic.